

Question 16**1 mark**

Identify the range of the function $g(x) = \frac{1}{2}f(x+1)$, given that the range of the function $y = f(x)$ is $[-6, 4]$.

- a) $[-12, 8]$
- b) $[-7, 3]$
- c) $[-5, 5]$
- d) $[-3, 2]$

Question 17**1 mark**

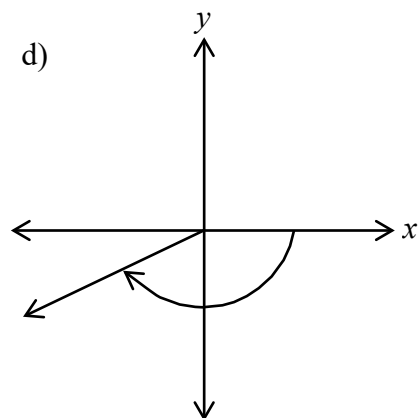
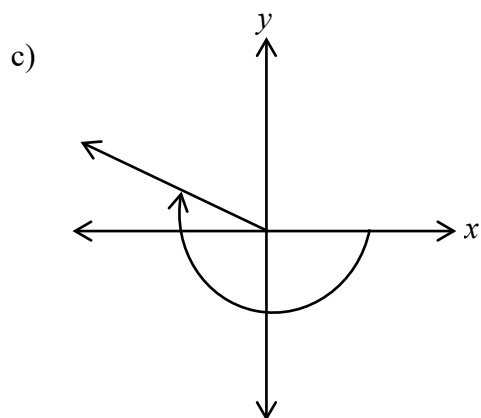
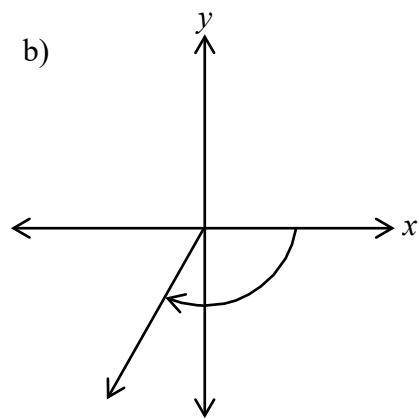
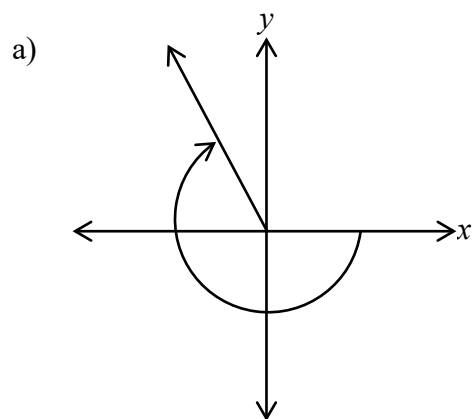
Identify the value of a , given that there are 11 terms in the expansion of $(3x^4 - y)^{2a}$.

- a) 5
- b) 6
- c) 10
- d) 11

Question 18

1 mark

Identify the angle that best represents $\theta = -\frac{6\pi}{5}$.

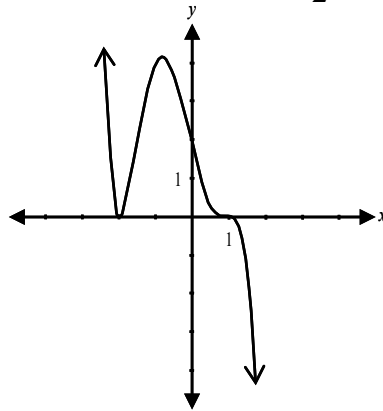


Question 19

1 mark

Identify a possible value for n , given the graph of $y = -\frac{1}{2}(x+2)^2(x-1)^n$.

- a) 1
- b) 2
- c) 3
- d) 4



Question 20

1 mark

Identify the statement that is false, given $g(x) = \frac{8x^2}{x^2 - 16}$.

- a) the graph of $g(x)$ has one x -intercept.
- b) the graph of $g(x)$ has a point of discontinuity (hole) at $x = 0$.
- c) the graph of $g(x)$ has two vertical asymptotes.
- d) the graph of $g(x)$ has a horizontal asymptote at $y = 8$.

Question 21

1 mark

Identify the equivalent form of $\log_a \left(\frac{1}{x^2} \right)$.

- a) $-2\log_a x$
- b) $1 - 2\log_a x$
- c) $2\log_a x$
- d) $-2\log_a \left(\frac{1}{x} \right)$

Question 22**1 mark**

Identify which one of the following expressions is equivalent to ${}_{13}C_6$.

a) ${}_{13}P_6$

b) ${}_{13}C_7$

c) ${}_{12}P_7$

d) ${}_{12}C_6$

Question 23**1 mark**

Identify the equation of $h(x) = f(x) - g(x)$, given $f(x) = x + 5$ and $g(x) = 4x + 1$.

a) $h(x) = -3x + 6$

b) $h(x) = -3x + 4$

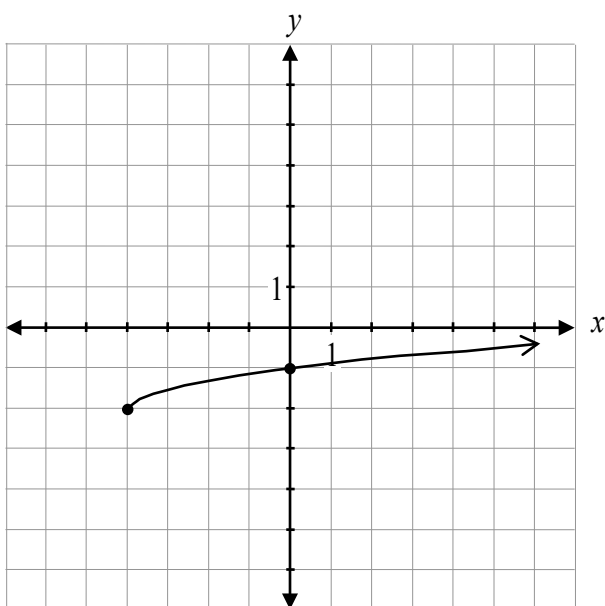
c) $h(x) = 3x + 6$

d) $h(x) = 3x - 4$

Question 24

3 marks 116

Determine the equation of the radical function represented by the graph.



$y =$ _____

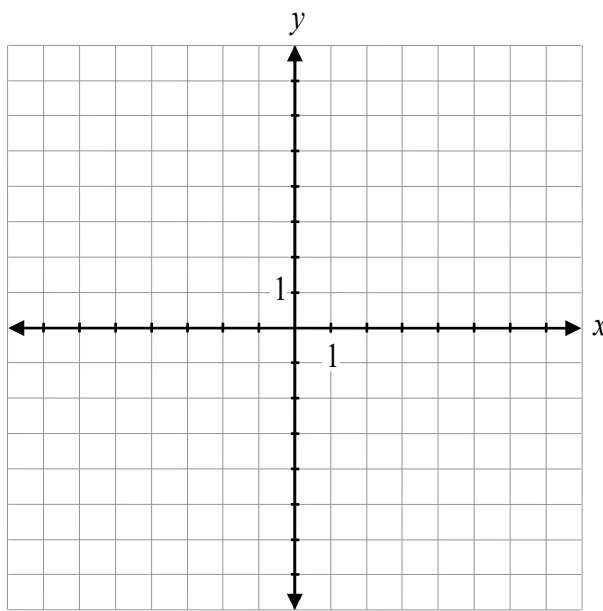
Determine the exact value of x .

$$\sec\left(\frac{2\pi}{3}\right)\left(\sin\left(-\frac{5\pi}{3}\right)\right)(x) = 3$$

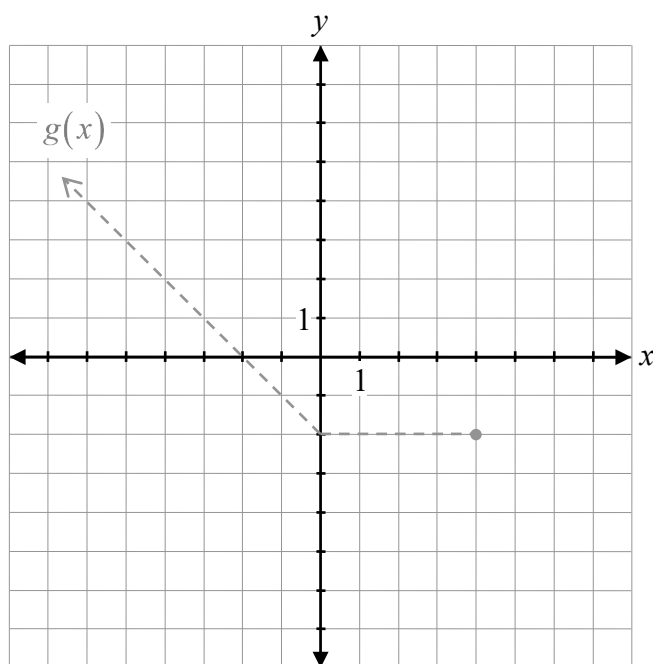
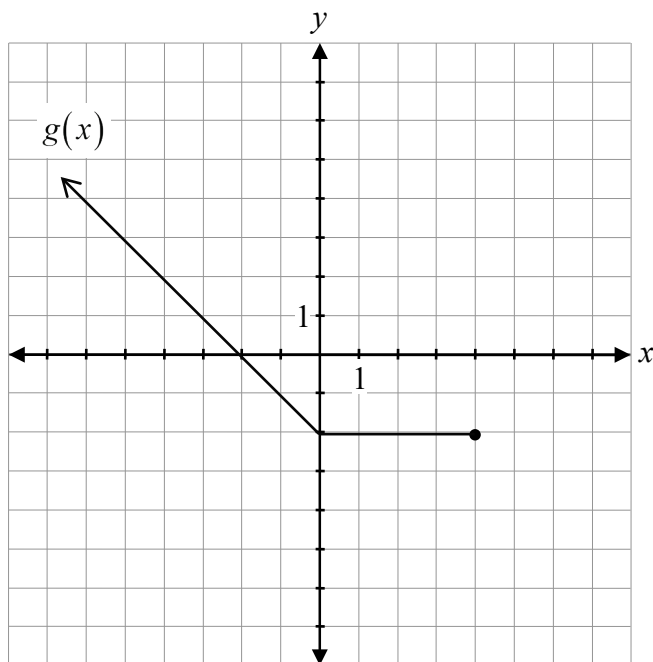
Question 26

3 marks 118

Sketch the graph of $y = 2^{-x} - 3$.



Given the graph of $y = g(x)$, sketch the graph of $y = \frac{1}{g(x)}$.



The graph of $g(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $g(x)$.

Determine the exact value of $\tan\left(\frac{\pi}{12}\right)$.

Explain why the graph of $g(x) = \frac{3}{x^2 + 4}$ does not have a vertical asymptote.

Question 30

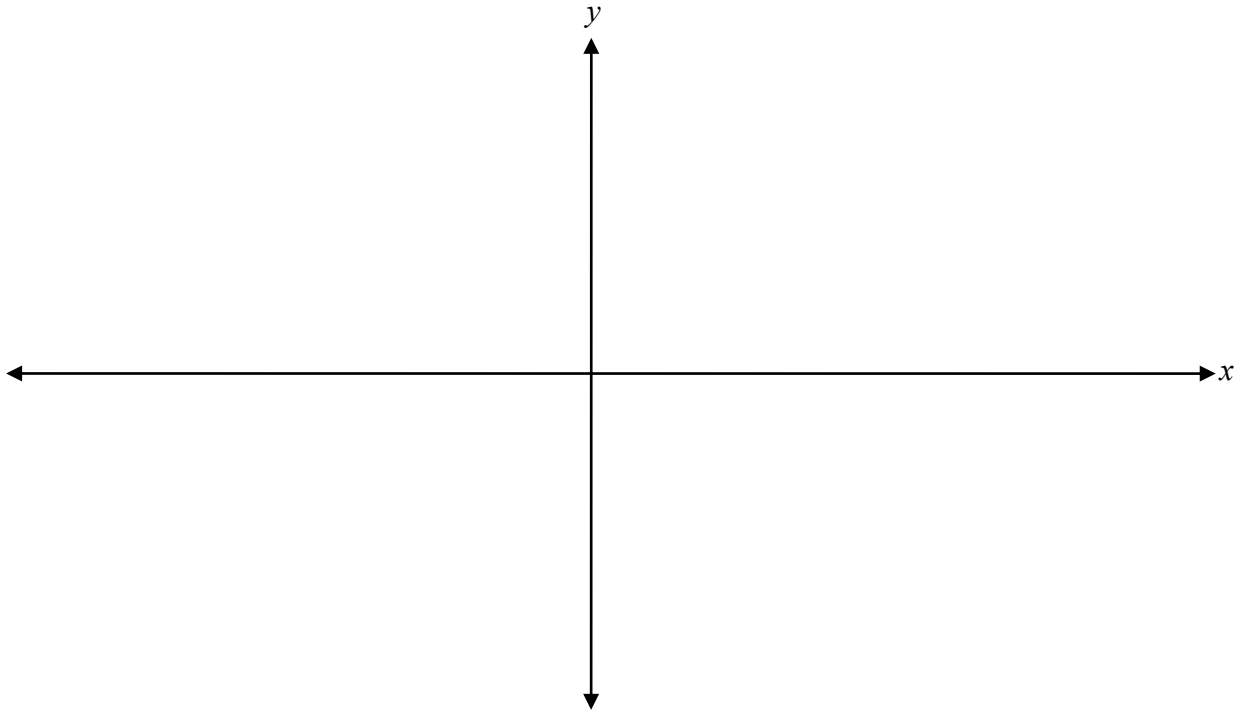
3 marks

122

Solve, algebraically.

$$\log_3 x + \log_3 (x + 8) = 2$$

Sketch at least one period of the graph of the function $y = \sin(3(x + 30^\circ)) - 1$.

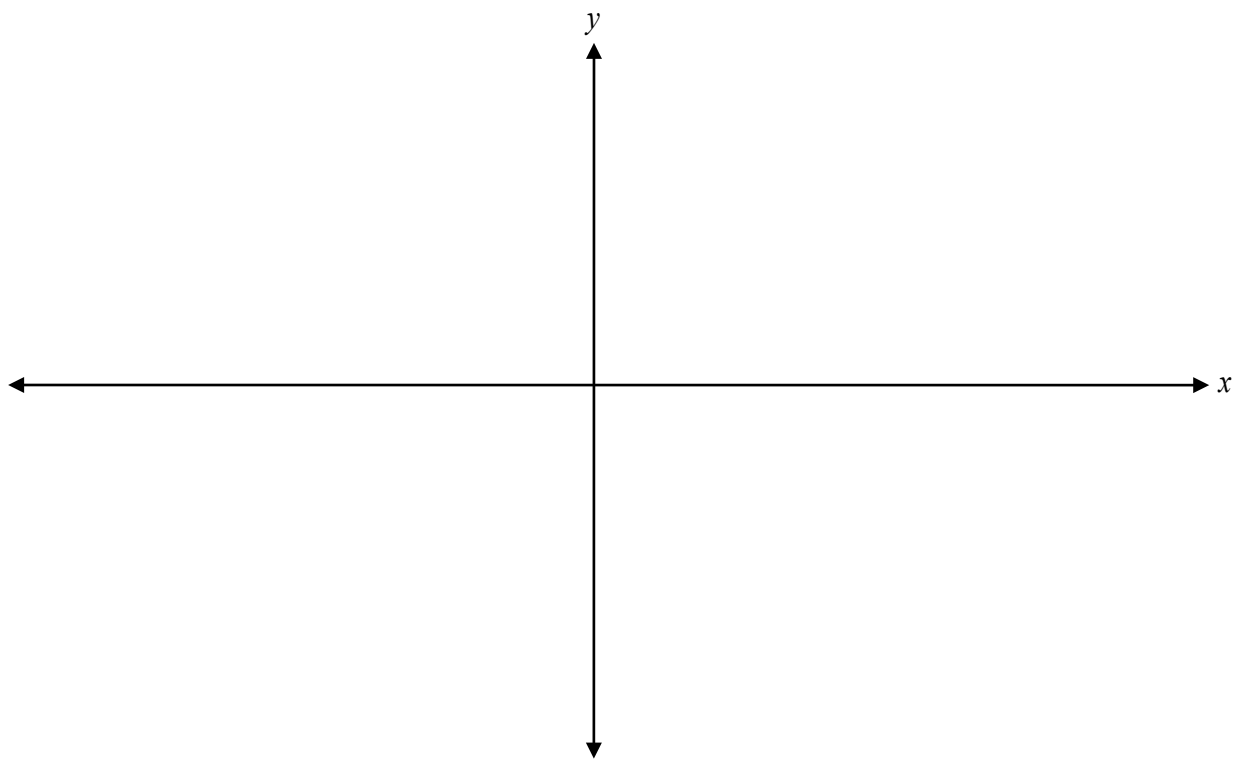


Question 32

1 mark 124

Explain why the domain of the function, $f(x) = \log(x - 3)$, is $x > 3$.

Sketch the graph of $p(x) = -(x-3)(x+1)^2(x-5)$.



Question 34

2 marks 126

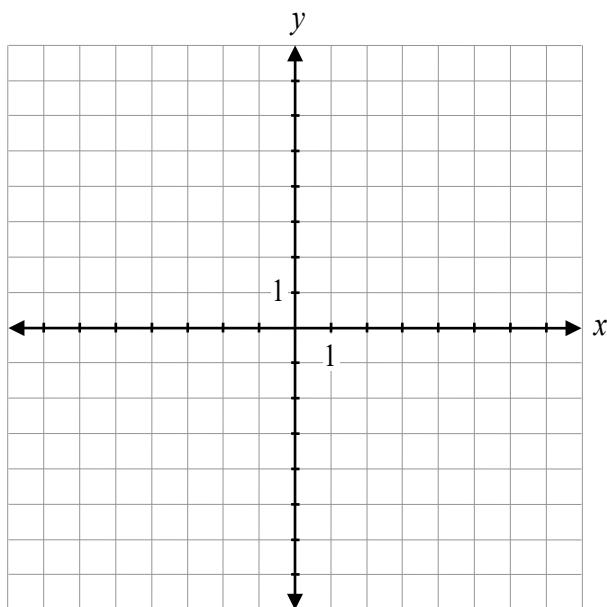
Given that $\sin \theta = -\frac{2}{3}$ and $\tan \theta > 0$, determine the exact value of $\sin 2\theta$.

Question 35

1 mark 127

Justify whether $\frac{5\pi}{8}$ and $-\frac{11\pi}{4}$ are coterminal angles.

Sketch the graph of $f(x) = \frac{-2x(x+1)(x-3)}{2x}$.



Given $\frac{\sin \theta + \cos \theta \csc \theta}{\sin \theta}$, determine the non-permissible values of θ , where $\theta \in \mathbb{R}$.

Question 38

2 marks 130

Write an equation of a rational function that has a horizontal asymptote at $y = 0$ and a vertical asymptote at $x = 6$.

Question 39

a) 1 mark b) 2 marks

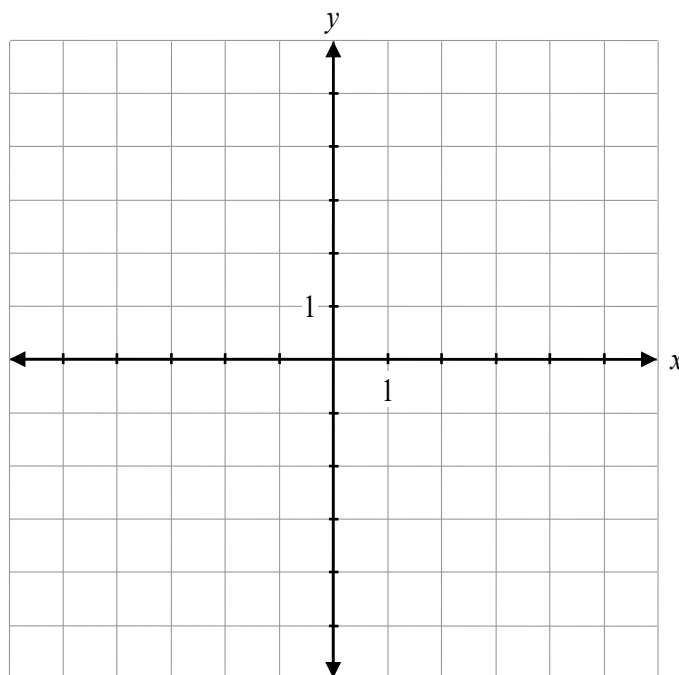
131
132

Given the functions $f(x) = \sqrt{x-1}$ and $g(x) = x^2$,

a) state the equation of $g(f(x))$.

$g(f(x)) =$ _____

b) sketch the graph of $g(f(x))$.



Question 40

1 mark 133

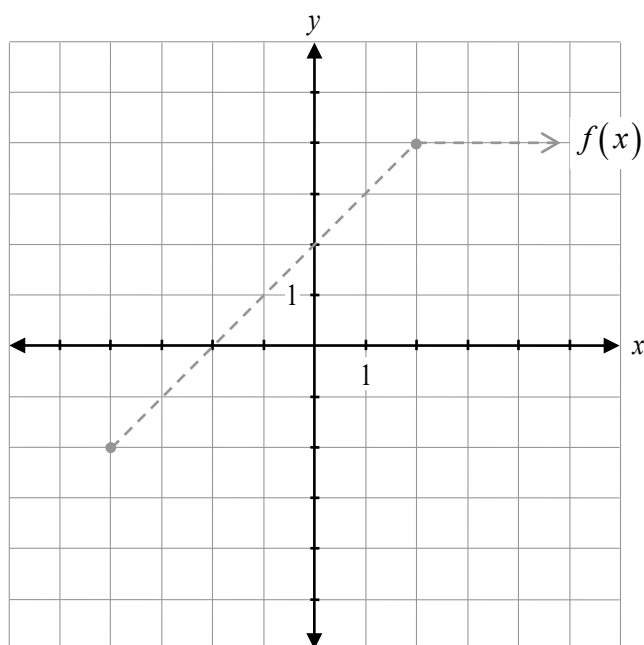
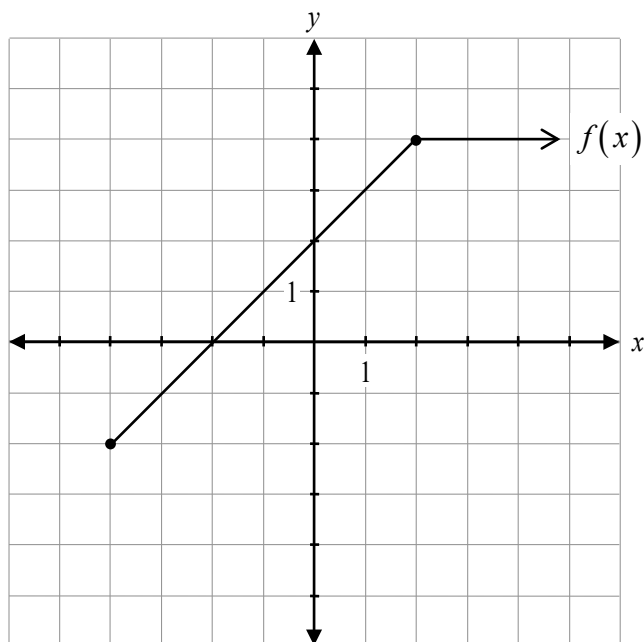
Suzanne was asked to determine the value of $\tan \theta$, given that $\sec \theta = -\frac{8}{3}$ and θ terminates in quadrant II.

Her solution:

$$\begin{aligned}(-3)^2 + y^2 &= (8)^2 \\ y^2 &= 55 \\ y &= \sqrt{55} \\ \tan \theta &= \frac{\sqrt{55}}{3}\end{aligned}$$

Describe her error.

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference. No marks will be awarded for the graph of $f(x)$.

The point $P(\theta) = (0, -1)$ lies on the unit circle. State the angle θ , over the interval $[2\pi, 4\pi]$.

Question 43

1 mark 136

Describe how the transformations of $f(x)$ on the graphs of $g(x) = f(3x - 6)$ and $h(x) = f(3(x - 6))$ are different.

Question 44

a) 1 mark b) 1 mark

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138

a) Solve.

$$\sqrt{2x+5}-3=0$$

b) Describe how the solution in a) relates to the graph of $y = \sqrt{2x+5} - 3$.

Determine all of the zeros of the function $p(x) = x^3 - 2x^2 - 9x + 18$.

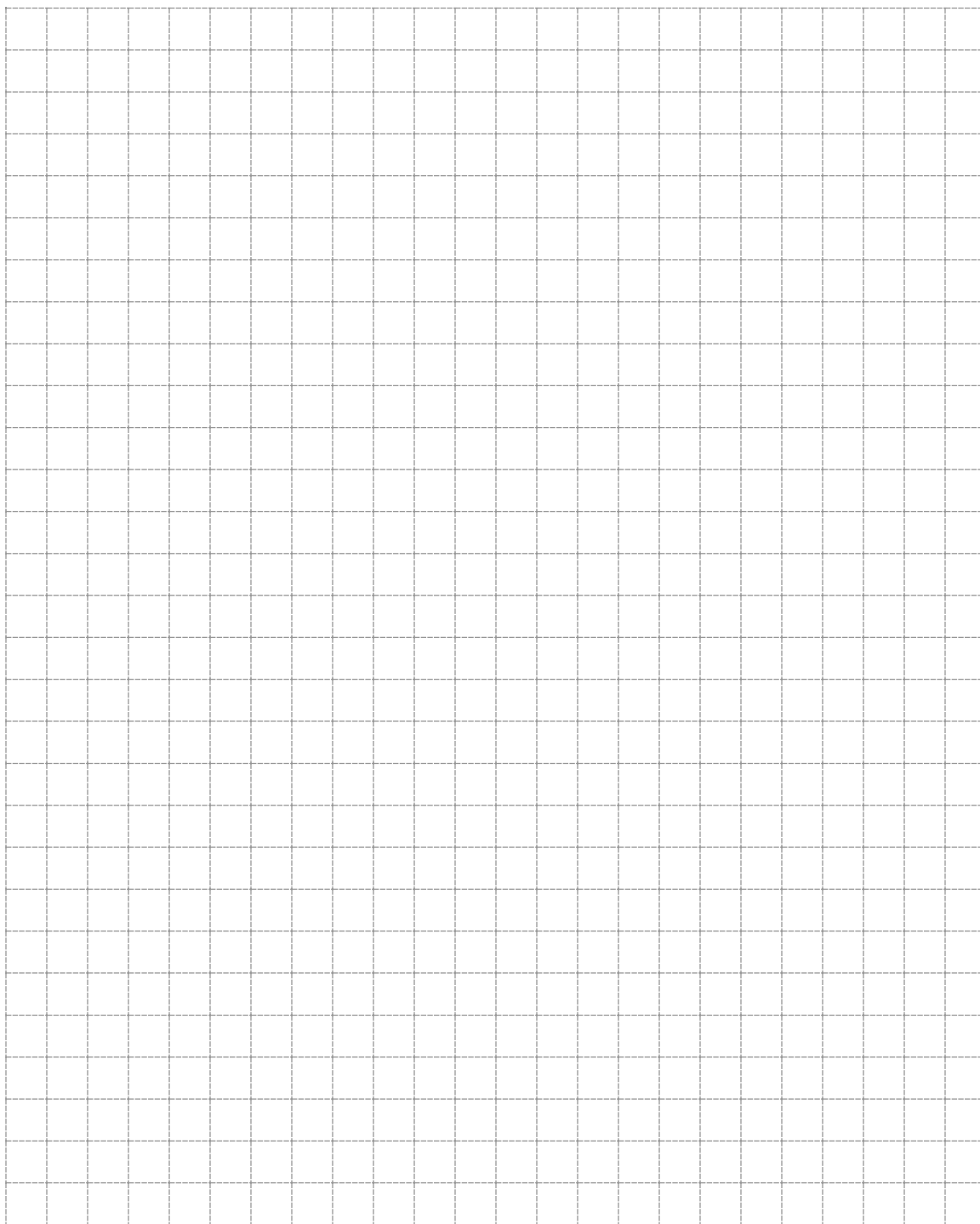
Given that the point $\left(\frac{\sqrt{23}}{6}, y\right)$ is on the unit circle, determine the exact value(s) of y .

Question 47

1 mark 141

State one zero of the function $y = \tan x$.

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.